

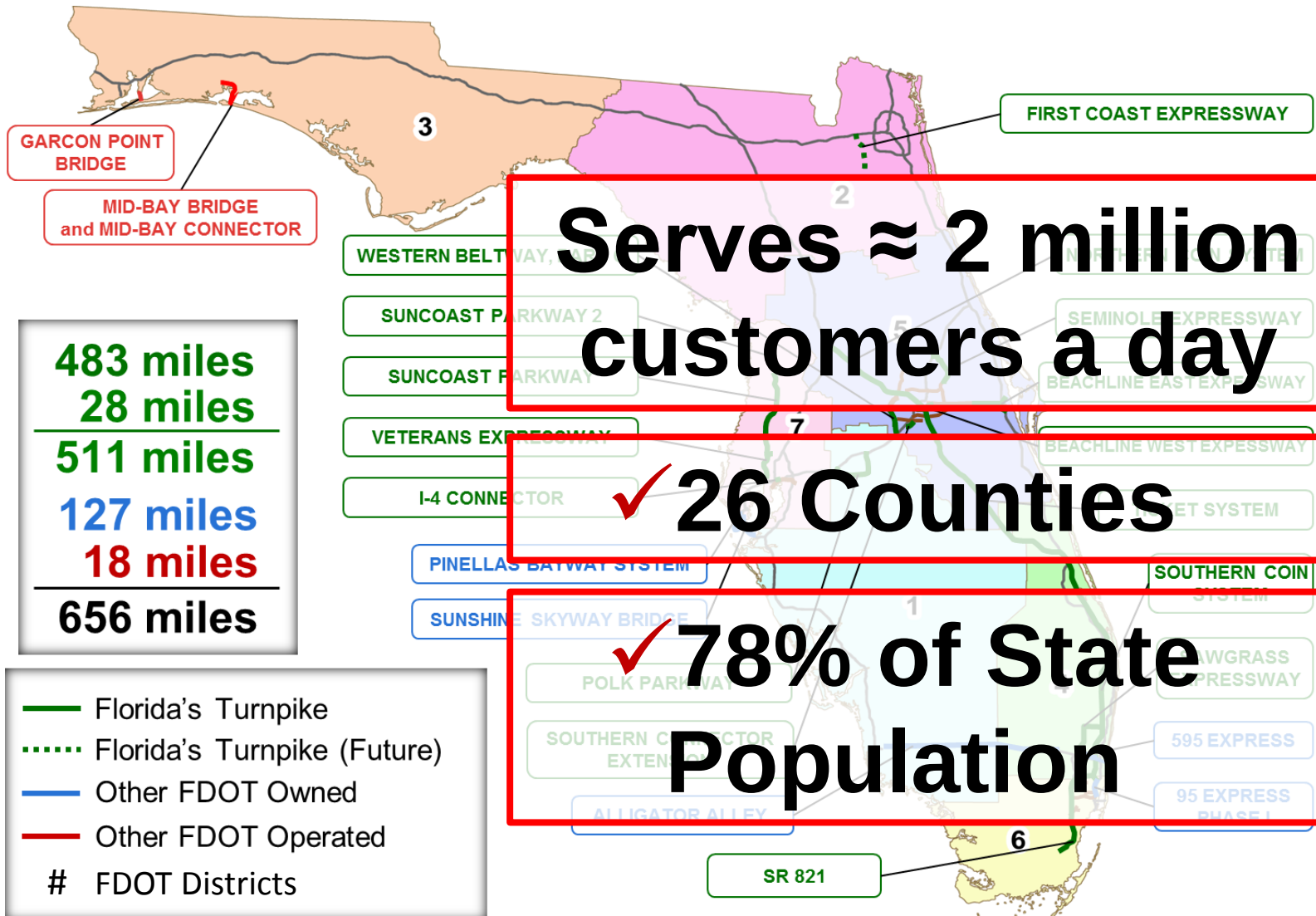


Florida Department of
TRANSPORTATION

Aligning Big Data Management with Business Priorities

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Florida's Turnpike Enterprise
Technology Project Manager

Turnpike System Map



Study Rationale

Question: Why was this study undertaken?

Answer: **Business Drivers & Recurring Themes**

Business Drivers

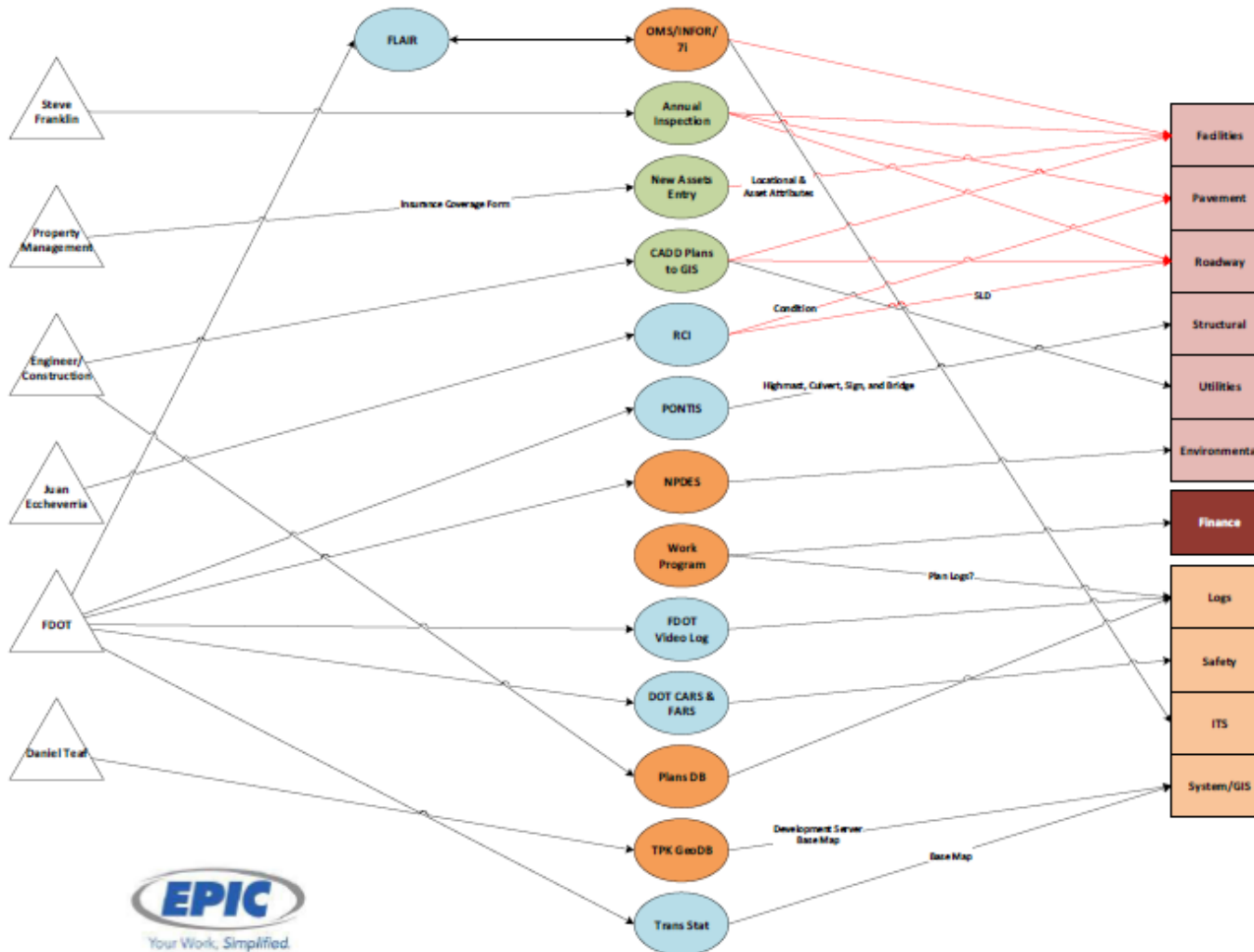
- **Compliance with Bond Requirements**
 - Investor Confidence
 - Excellent Rating
- **Program Level Funding Development**
 - Where, When and How much \$\$\$
- **Improving Data and Asset Management**
 - GASB34 Std. Compliance

State of the Data

Data Owners

Data Sources

User Interface



Recurring Themes

- **Multiple Data Sources**
- **Confidence in the Data**
- **Issues with Data Access**
- **Undefined “Sources of Truth”**

Case Study Overview

Assessment & Optimization of FTE's Asset Data Collection and Management Processes

- Stage 1: Preliminary Assessment (3 months)
 - Inventory
 - High-level recommendations
- Stage 2: Detailed Assessment (6 months)
 - Document priority data systems
 - Business processes mapping
 - Solution recommendations

Case Study Overview

Project Leaders

- FDOT/FTE Staff



- FTE General Engineering Consultants (GECs)



- FTE Consultant



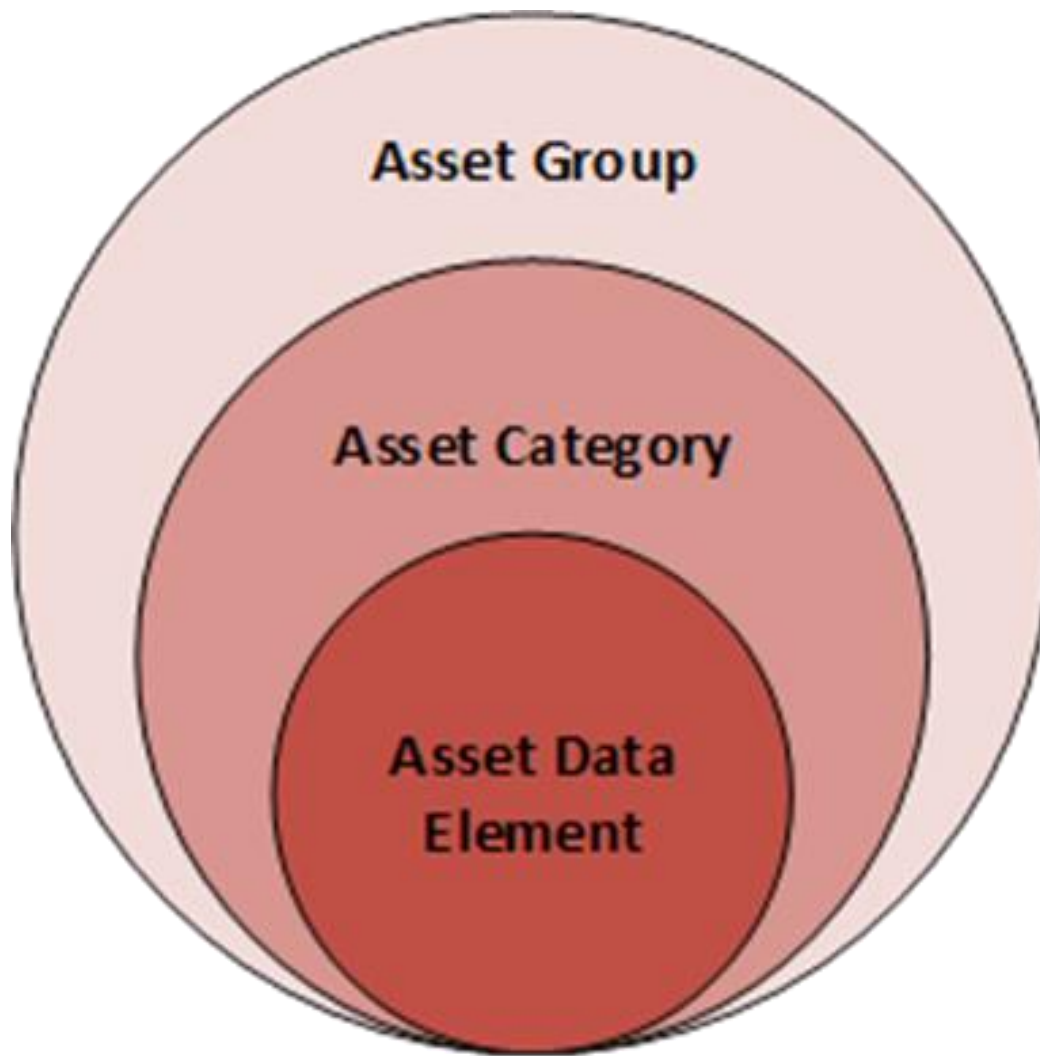
Key Project Program Areas/Units

- Finance
- Design
- Engineering
- Toll Systems
- Transportation Operations
- ITS (Intelligent Transportation System)
- Facilities/Roadway Maintenance

Project Methodology

- **Program Area /Unit Group Interviews**
 - Collecting information on current/as-is state:
 - Data sets used
 - Data systems used
 - Data management processes (creation, updates, etc.)
 - Collecting information for future/to-be state:
 - Data needs
 - Data systems needs (functionality)
 - Data management process improvements

Project Methodology



Facilities

Buildings

Generators,
Elevators, Fuel
Tanks, Fire
Extinguishers

Findings & Responses

Challenges Encountered:

- What's in it for me?
- Skepticism of a “top-down mandate”
- We know what needs to be done and how to do it
- We have no challenges

Response to Challenges:

- Clearly communicate project goals
- Establish alignment between data/system needs and business needs
- Explain the benefits of reliable and accessible data
- Engage end-users to increase solution acceptance

Recommendations

Project Recommendations:

- **Big Data needs to be GOOD Data**
- **Big Data needs to be ACCESSIBLE**
- **Big Data needs to align to BUSINESS NEEDS**



Recommendations

Big Data needs to be GOOD Data

- Eliminate redundancy
 - creation
 - capture
 - storage
- **Solution:** Establish “Source of Truth” for key datasets

Documenting Sources of Truth

FTE Asset-Level Data Collection, Storage, and Reporting Matrix

FTE Asset-Level Data Collection, Storage, and Reporting Matrix										Pro		S		as		*
Asset Category	Asset Data Element	AMS	MS	AIR	NTIS	I	ntal Office Pavement	Annual Inspection	FTE Renewal & Replacement	O & M Contractors	FTE Maintenance (Facilities)	FTE Maintenance (Roadway)	FTE Traffic Operations (ITS)	FTE Tolls	FTE Environmental Management Office	No Clearly Identifiable Source of Truth
Pavement	Pavement Depreciation															
Pavement	Pavement Condition							T								
Pavement	Pavement Depreciation							T								
Facilities	Building Assets	1000KW Generator	S	T	S					D	D		R			T

D = Asset data is collected by assigned workgroup or data collection activity

S = Data Storage location for the asset data element

T = *Source of Truth* for the asset data element

R = Reporting is performed by the assigned workgroup on the asset data element

Recommendations

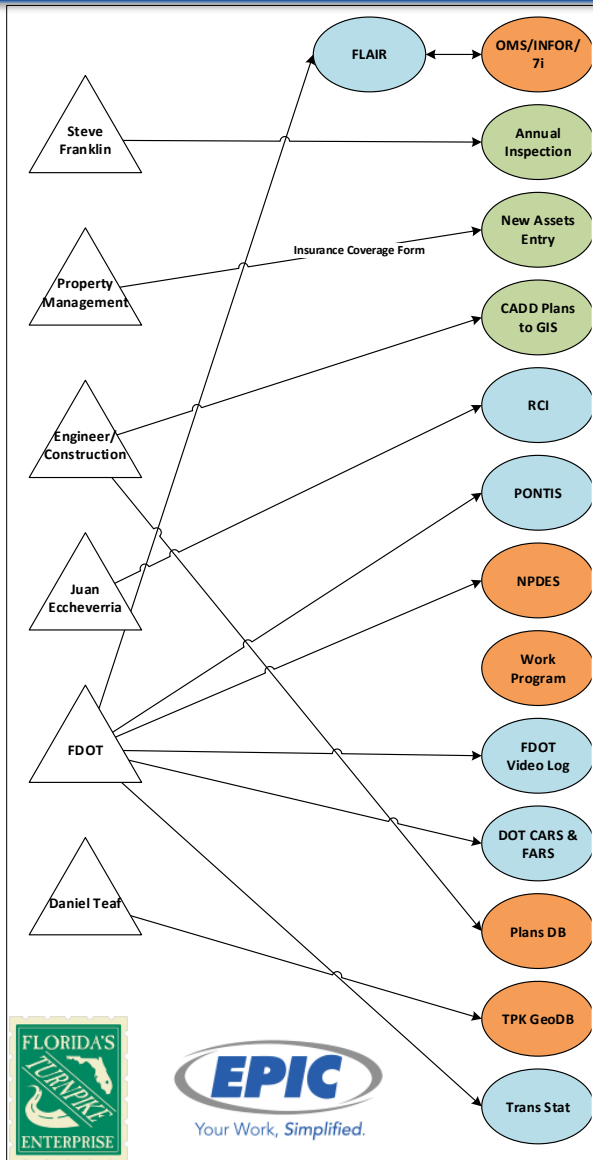
Big Data needs to be ACCESSIBLE

- Easy to Find
- Easy to Use
- Map-based Interface

Solution: Enterprise Data Repository (EDR)

- Everyone sees same data (as much as needed)
- Everyone accesses data from same system
- Leverages FDOT GEV (GIS) framework

Proposed EDR Concept



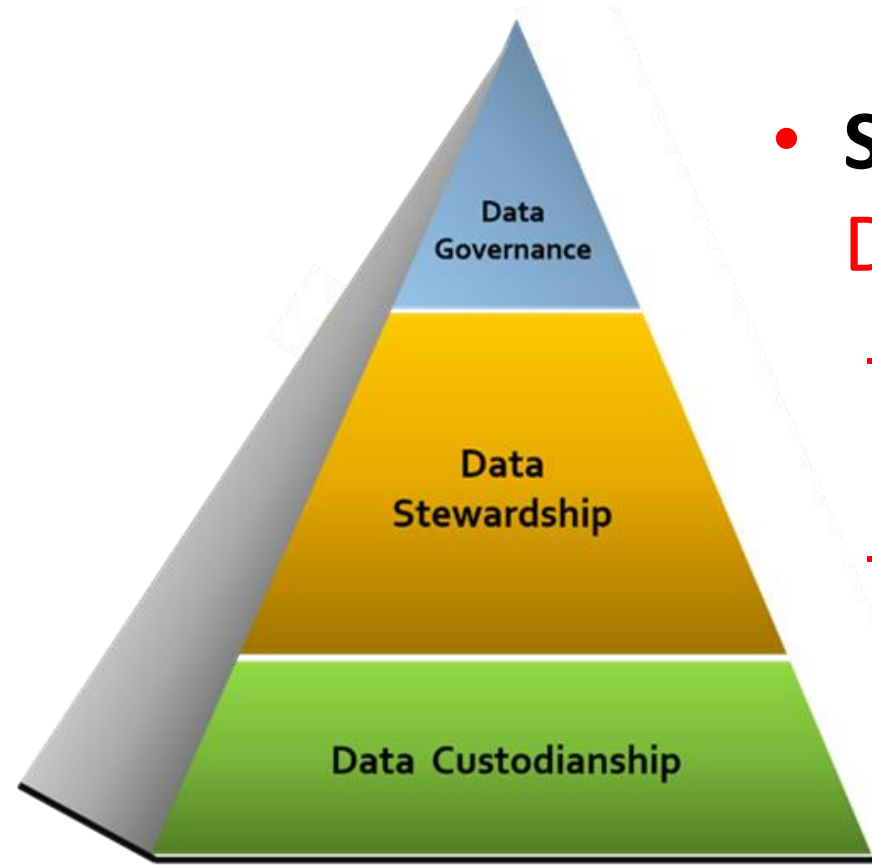
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Recommendations

Big Data needs to align to BUSINESS NEEDS

- Identify Priority Areas
 - Modular Approach
 - Crawl, Walk, Run
 - Business Priorities Drive Solution
- Support decision making
 - Spatially integrated data visualization
 - Better business decisions and improved program funding

Recommendations



- **Solution:** Establish Asset Data Governance Plan
 - Focused on Turnpike Asset Data
 - Aligned with FDOT's ROADS Initiative (Reliable, Organized and Accurate Data Sharing)

Collaboration

- Communication with FDOT's Business Systems Support Office (BSSO)
- Shares ROADS Objectives
- Leverage existing FDOT resources
 - GEV framework
- Open Data
 - FTE-specific data
 - District-specific data
 - Shared Data
- Shared solution

Closing Thoughts

- FTE believes in Big Data Analytics
- Big Data management requires strategic vision
- Support is required from executive level
- Big Data is not a one-time investment
- Big Data initiatives must be seen as *business priorities and not IT priorities*

Closing Thoughts

***“Vision of the Future - Better Data,
Smarter Decisions”
– Jim Boxold***